# Computer Circuit

Condensed Study Notes

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## An Introduction to Computer Circuit

Computer circuits are the internal pathways that dictate how fast a computer operates. Speed is increased by reducing the time it takes for switches to open and close, and by designing circuit routes that can handle faster electron flow.

In a theoretical sense, a computer circuit is a model for computation. Think of it like a series of steps where input values are processed through different components, called gates, each performing a specific function. This is a generalization of Boolean circuits and a mathematical model for digital logic circuits.

These circuits are defined by the types of gates they contain and the values these gates can produce. For example, in a Boolean circuit, the values are either true or false (Boolean values), and the gates perform operations like conjunction (AND), disjunction (OR), and negation (NOT).

For a computer, a circuit essentially forms a complete path, or a network of connected paths, that allows electrons to flow. The core concept of a computer circuit is binary, meaning it has only two possible states: on or off. This is achieved using transistors, which act like tiny, super-fast on-off switches that can be electronically controlled in incredibly short periods – nanoseconds (billionths of a second) and picoseconds (trillionths of a second).

## Motherboard as the main circuit board in a PC

The motherboard is the primary printed circuit board (PCB) within a computer. Think of it as the central hub or backbone that connects all the different parts of the computer together, allowing them to communicate.

Virtually all computers, especially desktops and laptops, have a motherboard. It serves as the main point of connectivity for both internal components and external peripherals.

Internal components that connect through the motherboard include:

- Chipsets
- The Central Processing Unit (CPU) – the computer's brain.
- Memory (RAM)

External peripherals that connect to the motherboard often include:

- Wi-Fi cards
- Ethernet ports for wired internet
- Graphics cards, which house the Graphics Processing Unit (GPU) for visual output.

## IC as the major Circuitry Technology in Modern PC

Integrated circuits (ICs), also known as microchips, are fundamental to modern electronics. Think of them as incredibly complex miniature city blocks, packed with tiny components like transistors, resistors, and condensers, all working together.

These ICs are designed to perform specific, often complex, tasks in a very compact and efficient way. They are built on a silicon wafer and have multiple terminals to connect to other parts of a system, allowing them to carry out various functions. They are found everywhere, from the memory chips in your computer to microcontrollers in automobiles.

Key points:

- ICs are miniature devices containing many electronic components like transistors and resistors.
- They are designed for specific tasks and come in various shapes and sizes.
- They are crucial for performing complex functions in a compact space.

## What is an Integrated Circuit

An integrated circuit (IC), also known as a microchip, is a miniaturized electronic circuit containing multiple electronic components. These components, such as MOSFETs (Metal Oxide Semiconductor Field Effect Transistors), are all built together on a single, flat piece of semiconductor material.

Think of it like a tiny, self-contained city of electronic parts, all working together seamlessly. This close integration makes ICs faster, more reliable, more efficient, and less expensive than circuits built with individual components.

Based on their construction, integrated circuits are typically divided into three main types:

- **Digital Integrated Circuits:** These handle signals that are either on or off, representing binary data (0s and 1s).
- **Analog Integrated Circuits:** These work with signals that vary continuously over a range, like audio or temperature.
- **Mixed Integrated Circuits:** These combine both digital and analog components on the same chip.